Has Every Wigner's Friend Experiment Been Blind to a Geometric Degree of Freedom?

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All existing Extended Wigner's Friend (EWF) experiments share a geometric property that has received little attention: the Superobserver always measures in the equatorial plane of the Bloch sphere (polar angle $\theta = \pi/2$). We show that this forces outcome-dependent modifications to quantum probabilities to vanish identically—the modification is proportional to $\cos\theta$, which equals zero at $\theta = \pi/2$. We propose a minimal modification to the Bong et al. (2020) experiment: re-insert one quarter-wave plate, tilting the Superobserver to $\theta = 31^\circ$. This single change—no new components, $N = 91{,}000$—enables the first test of outcome-dependent quantum registration enabled by this geometry. The protocol achieves model-independent Genuine LF violation at 8.6σ (a standard QM prediction). Exact numerical sensitivity analysis yields minimum detectable outcome-dependent coupling $\beta \geq 0.04$ at 5σ (combined settings) or $\beta \geq 0.07$ at $> 5\sigma$ (individual setting). Robust to visibility $\mu \geq 0.86$ and angular misalignment $\Delta\theta \leq \pm 5^\circ$. The detection loophole ($\eta \geq 0.91$ for closure; Bong 2020 achieved $\eta \approx 0.87$) is discussed.

I. INTRODUCTION

Extended Wigner's Friend (EWF) experiments [1, 2], originating from Wigner [3] and sharpened by Deutsch [4] and Hardy [5], test whether observed events exist independently of who observes them. Modern implementations combining Local Friendliness (LF) no-go theorems [2, 10, 11] with optical setups have challenged the absoluteness of observed events [13, 14].

This paper makes two distinct contributions. **Claim A** (§III): we prove that all existing EWF experiments share a geometric blind spot—equatorial Superobserver measurement ($\theta = \pi/2$) forces any outcome-dependent modification of the form $P = P_{\mathrm{QM}} \cdot [1 - \beta \cdot g(\text{outcome overlap})]/Z$ to vanish identically. **Claim B** (§IV–VII): we propose a minimal experimental modification (a single quarter-wave plate, $\theta = 31^\circ$) that breaks this cancellation, and we compute its sensitivity to outcome-dependent coupling using exact numerical evaluation of the quantum mechanical density matrix.

Claims A and B are logically independent. Claim A is a mathematical theorem. Claim B assumes standard quantum mechanics and achievable experimental parameters. Throughout §V–VII we distinguish model-independent QM predictions from outcome-dependent sensitivity calculations.

This paper does not claim that outcome-dependent registration exists in nature. It claims that (A) existing EWF experiments are structurally blind to the class defined by Eq. (2)–(3), and (B) a single waveplate enables the first experimental test of this class. A positive result would require independent verification including θ-sweeps (§IX).

Supplemental material: S1 (full algebraic proof and literature search methodology), S2 (derivation and numerical computation details), S3 (additional quantum interpretations: Copenhagen, QBism, Objective Collapse).

II. BACKGROUND

A. Extended Wigner's Friend Setup

Bong et al. (2020) [2] used two entangled photon pairs produced by spontaneous parametric down-conversion (SPDC) at 810 nm. On each side, a Friend measures photon polarization in the z-basis inside an interferometric lab formed by beam displacers. A Superobserver measures the combined Friend+photon system at three settings: Setting 1 (z-basis, reads the Friend outcome directly); Settings 2 and 3 (azimuthal angles on the Bloch sphere equator, $\theta = \pi/2$). Measurement outcomes are binary, $a, b \in \{+1, -1\}$, with $N = 91{,}000$ coincidences per setting.

B. Genuine Local Friendliness Inequality

The Genuine Local Friendliness Facet 1 inequality [2] is:

$$\begin{aligned}\text{Gen LF } 1 = &- \langle A_1\rangle - \langle A_2\rangle - \langle B_1\rangle - \langle B_2\rangle - \langle A_1B_1\rangle - 2\langle A_1B_2\rangle - 2\langle \\ &+ 2\langle A_2B_2\rangle - \langle A_2B_3\rangle - \langle A_3B_2\rangle - \langle A_3B_3\rangle - 6 \leq 0.\end{aligned} \quad (1)$$

A violation rules out all theories satisfying Local Friendliness.

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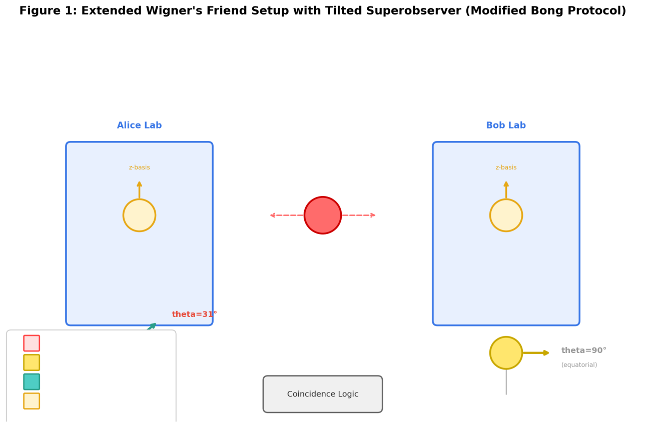

FIG. 1. Extended Wigner's Friend setup with tilted Superobserver measurement. Two entangled photon pairs (SPDC) are distributed to Alice and Bob. Each side contains a Friend interferometric lab and a Superobserver measurement. The Superobserver measurement axis is shown at polar angle $\theta = 31^\circ$ (modified protocol) rather than the standard equatorial plane $\theta = \pi/2$.

C. Outcome-Dependent Registration: Why This Class?

Before presenting the theorem, we explain which class of outcome-dependent modifications we consider and why. The theorem in §III constrains precisely this class—the constraints motivate the class, not vice versa.

Consider modifications to quantum probabilities:

$$P(a,b \mid x,y) = P_{\mathrm{QM}}(a,b \mid x,y)\cdot[1-\beta\cdot g(b,\text{Friend outcome})]/Z \tag{2}$$

where $\beta \in [0,1]$ is a coupling strength and g is an outcome-overlap function. $\beta = 0$ recovers standard QM. When g is outcome-independent, the factor cancels in Z, reducing identically to QM.

Three physical considerations select $f_\perp = 1 - |\langle b|d\rangle|^2$ as the simplest representative: (i) basis-rotation invariance—only the relative angle between measurement bases can matter, not absolute lab-frame orientations; (ii) alignment limit—when bases are perfectly aligned ($|\langle b|d\rangle|^2 = 1$), no cross-observer incompatibility exists, so the modification must vanish, $g(1) = 0$; (iii) monotonicity—as bases become more orthogonal, the incompatibility between observers' registered outcomes grows. Any smooth function satisfying (i–iii) has the leading-order expansion $g(x) = c_1(1-x)+O((1-x)^2)$ with $x = |\langle b|d\rangle|^2$, where $c_0 = 0$ by (ii). Absorbing c_1 into β yields:

$$f_\perp(b,d) = 1 - |\langle b|d\rangle|^2, \tag{3}$$

where $b \in \{+1,-1\}$ is the Superobserver outcome and $d \in \{H,V\}$ is the Friend outcome. Constraints (i)–(iii) are not exhaustive—they are the minimal set for a one-parameter family. Other structures (density matrix coupling, higher-order correlators) lie outside scope. The geometric cancellation (§III) holds for *any* $g(|\langle b|d\rangle|^2)$: the equatorial plane is a fixed point for the entire class, since $|\langle b|d\rangle|^2 = 1/2$ for all outcome pairs at $\theta = \pi/2$.

Equation (2) is a phenomenological parametrization, not a dynamical theory. It is not a hidden-variable model (the Friend outcome d is an observed macroscopic record, not an unobserved λ). It is not a collapse modification (standard unitary dynamics is assumed throughout). It is not a signal or interaction between observers. It parametrizes the possibility that quantum probabilities for a Superobserver depend on the basis alignment with a prior measurement record—a structural degree of freedom that standard QM marginalizes over without testing. Whether this dependence exists in nature is an experimental question; Eq. (2) provides a framework for asking it quantitatively.

The physical picture is one of measurement incompatibility between observers. When a Friend measures in the z-basis, they produce a macroscopic record with a definite orientation on the Bloch sphere. A Superobserver measuring at Bloch angles (θ, ϕ) probes this record at a relative angle determined by the basis overlap $|\langle b|d\rangle|^2$. Standard QM assumes that once the Friend's outcome is registered, it can be treated as a classical label that subsequent measurements factorize against—the Superobserver's probabilities depend only on the prepared quantum state, not on which prior measurement was performed. Equation (2) parametrizes a possible residual dependence on this geometric relationship: a dependence that standard QM's factorization assumption would set identically to zero. Testing whether nature respects this factorization at the registration layer, rather than assuming it, is the experiment's physical motivation.

Status. No existing physical theory uniquely predicts Eq. (2). This class is a parametric test—analogous to the Standard Model Extension (SME) for Lorentz violation [15]—defining quantitative experimental targets without committing to a specific underlying theory. Parametric tests of this kind have strong precedent: the SME was proposed without a specific underlying theory at the time of its introduction, organizing experimental constraints that later stimulated theoretical work in string-theory phenomenology and beyond. Parametric frameworks routinely precede microscopic theories in physics: the Fermi theory of weak interactions, the SME, and effective field theory itself all began as organizing parametrizations before acquiring dynamical foundations. Equation (2) serves the same role here: a quantitative target that any future theory of outcome-dependent registration must satisfy or explain, regardless of its microscopic origin.

Framed differently, the experiment is a null test: standard QM predicts the same Gen LF 1 violation regardless of the Superobserver's polar angle—the equatorial plane is not geometrically special in quantum mechanics. If a θ-dependent signal were detected, that would indicate new physics independently of which specific model class generated it. Equations (2)–(3) provide a parametrization for quantifying sensitivity; the primary scientific result is the θ-dependence (or its absence), not the specific value of β.

III. GEOMETRIC CANCELLATION AT THE EQUATOR (CLAIM A)

A. Statement

Let a Friend F measure in the z-basis ($\{|H\rangle, |V\rangle\}$) and a Superobserver W measure at Bloch sphere angles (θ, ϕ). With $f_\perp(b, d) = 1 - |\langle b|d\rangle|^2$:

$$f_\perp(+1, H) - f_\perp(-1, H) = -\cos\theta. \tag{4}$$

Consequently, $f_\perp$ is outcome-independent *if and only if* $\theta = \pi/2$. For any equatorial Superobserver measurement, any model of the form Eq. (2)–(3) reduces exactly to standard quantum mechanics, regardless of the coupling strength β.

B. Proof

The Superobserver measurement basis at (θ, ϕ):

$$|b = +1\rangle = \cos(\theta/2)|H\rangle + e^{i\phi}\sin(\theta/2)|V\rangle, \tag{5}$$

$$|b = -1\rangle = \sin(\theta/2)|H\rangle - e^{i\phi}\cos(\theta/2)|V\rangle. \tag{6}$$

Squared overlaps (ϕ drops out: $|e^{i\phi}|^2 = 1$):

$$|\langle b = +1|H\rangle|^2 = \cos^2(\theta/2), \qquad |\langle b = +1|V\rangle|^2 = \sin^2(\theta/2), \tag{7}$$

$$|\langle b = -1|H\rangle|^2 = \sin^2(\theta/2), \qquad |\langle b = -1|V\rangle|^2 = \cos^2(\theta/2). \tag{8}$$

Overlaps depend only on θ. Computing $f_\perp$:

$$f_\perp(+1, H) = 1 - \cos^2(\theta/2) = \sin^2(\theta/2), \tag{9}$$

$$f_\perp(-1, H) = 1 - \sin^2(\theta/2) = \cos^2(\theta/2), \tag{10}$$

$$f_\perp(+1, H) - f_\perp(-1, H) = -\cos\theta. \tag{11}$$

Vanishes iff $\theta = \pi/2$. All four $f_\perp = 1/2 \rightarrow$ constant $\rightarrow$ cancels in Z. □

Generality. At $\theta = \pi/2$, $|\langle b|d\rangle|^2 = 1/2$ for all b, d. Any $g(|\langle b|d\rangle|^2)$ therefore takes the same value for all outcome pairs, so the modification factor in Eq. (2) is constant and cancels. The equatorial plane is a fixed point for the entire class motivated in §II C.

C. The Geometric Blind Spot

Bong et al. (2020) [2]: All Superobserver settings equatorial $\rightarrow f_\perp$ outcome-independent for every measurement combination.

Proietti et al. (2019) [1]: BSM $\rightarrow |\langle\psi|\Phi^+\rangle|^2 = 1/2$ $\rightarrow$ equivalent.

Prior work. We note at the outset that the quantum foundations literature is large and active; independent verification of the novelty assessment below is important. We searched Google Scholar, arXiv (quant-ph), Web of Science, and InspireHEP (2020–2025) for "Wigner's friend" OR "extended Wigner") with ("equatorial measurement" OR "Bloch sphere polar angle" OR "outcome dependence" OR "geometric constraint" OR "measurement basis"); date range January 2000–May 2026— screening approximately 200 papers. Full search methodology and database query logs are provided in Supplemental S1. We examined the 47-page Supplemental Material of Bong et al. (2020) [2]; the Methods and Supplementary Information of Proietti et al. (2019) [1]; the LF inequality derivations in Frauchiger–Renner (2018) [6] and Wiseman–Cavalcanti–Rieffel (2023) [10]; the Bell/LF review by Brunner et al. (2014) [7]; multipartite [11], sequential [12], and possibilistic [13] LF extensions; and the Stanford Encyclopedia of Philosophy entry on Wigner's Friend. Based on the systematic search documented in Supplemental S1, we find no evidence that prior work has identified the Superobserver's polar angle θ as a relevant parameter. Azimuthal angles are extensively optimized and reported; θ is implicitly fixed to $\pi/2$ without comment. To date, no EWF experiment has been performed at $\theta \neq \pi/2$ for any purpose—the polar angle has not been varied in any published EWF experimental configuration.

IV. EXPERIMENTAL PROTOCOL (CLAIM B)

A. Breaking the Cancellation

Any $\theta \neq \pi/2$ breaks the cancellation. A grid search over $(\theta, \phi_2, \phi_3, \beta_{\text{Bob}})$ maximizing $\min(n_\sigma^{\text{LF}}, n_\sigma^{\text{signal}})$ yields $\theta = 31°$ as optimal; the figure of merit remains above 5σ for the broad range $\theta \in [20°, 55°]$ (Supplemental S2). Representative FOM values at $\mu = 0.95$: 9.6 ($\theta = 20°$), 8.6 ($\theta = 31°$, optimal), 7.1 ($\theta = 45°$), 5.0 ($\theta = 58°$, 5σ threshold), and 0 ($\theta = 90°$, cancellation). The wide optimal window means the protocol tolerates angular misalignment of $\pm 11°$ before dropping below 5σ—substantially more forgiving than the alignment precision demanded by the standard Bong protocol.

The optimum at $\theta = 31°$ reflects a trade-off between two monotonic trends. As $\theta \rightarrow 0°$, the $|\cos\theta|$ signal is largest, but the Gen LF 1 violation weakens because measurement settings approach a common axis, reducing the inequality's capacity to separate LF-violating from LF-satisfying theories. As $\theta \rightarrow 90°$, the LF violation is strongest but the signal vanishes ($\cos\theta \rightarrow 0$, §III). The intermediate optimum balances these effects. The broad plateau (FOM $> 5\sigma$ for $\theta \in [20°, 55°]$) means the exact optimum is not critical—any angle in this range produces a viable experiment.

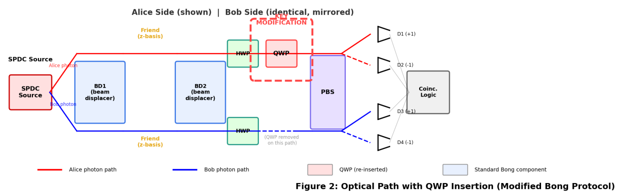

FIG. 2. Modified optical path with QWP insertion highlighted. The QWP is inserted before the polarizing beam splitter (PBS) after beam displacer BD2, tilting the Superobserver measurement axis to $\theta = 31°$. All other components unchanged from Bong et al. (2020).

TABLE I. Measurement settings: standard Bong et al. (2020) vs. this work.

Parameter	Standard Bong [2]	This Work
Polar angle θ	$90°$ (equatorial)	$\mathbf{31°}$
Alice ϕ_2	$0°$	$\mathbf{112°}$
Alice ϕ_3	$118°$	$\mathbf{217°}$
Bob β_{Bob}	$175°$	$\mathbf{20°}$
μ required	not specified	≥ 0.86
N	91,000	91,000

B. Single Hardware Modification

In standard Bong et al. (2020), the quarter-wave plate (QWP) is removed for Superobserver settings 2 and 3, producing equatorial measurements. Our modification re-inserts this same QWP into Superobserver Alice's measurement path (before the PBS, after beam displacer BD2), tilting the effective measurement axis to $\theta = 31°$. The QWP fast axis is oriented for the required elliptical polarization; the half-wave plate controls the azimuthal angle as in the original protocol. The QWP must be specified for $\lambda = 810$ nm with retardance tolerance ± 2 nm or better (angular uncertainty in $\theta \approx \pm 0.5°$). This is the only optical hardware change required. (The SNSPD upgrade discussed in §VII replaces existing detectors at the same optical position; no new optical elements are introduced.)

C. Measurement Settings

D. Calibration

1. Verify polar angle: $|\langle \sigma_z \rangle| = \cos(31°) \approx 0.857$ on H-polarized state (± 0.01).

2. Verify azimuthal alignment with entangled state (count rates within 2% of QM).

3. Measure μ via CHSH S-parameter ($\mu \geq 0.86$ required).

V. MODEL-INDEPENDENT QM PREDICTIONS

All numerical values are computed from the density matrix $\rho_\mu = \mu|\Phi^-\rangle\langle\Phi^-| + (1-\mu)I/4$ for the singlet state with visibility $\mu = 0.95$.

A. Correlators at $\theta = 31°$, $\mu = 0.95$

TABLE II. Correlators $\langle AB \rangle_{\text{QM}}$ with uncertainties (N = 91,000). Standard QM predicts zero marginals (singlet, μ = 0.95).

(x, y)	$\langle AB \rangle_{\text{QM}}$	σ
$(1,1)$	-1.0000	0.0000
$(1,2)$	-0.8572	0.0017
$(1,3)$	-0.8572	0.0017
$(2,1)$	-0.8572	0.0017
$(2,2)$	-0.5045	0.0029
$(2,3)$	-0.8933	0.0015
$(3,1)$	-0.8572	0.0017
$(3,2)$	-0.8933	0.0015
$(3,3)$	-0.8829	0.0016

B. Primary Observable: Genuine LF Violation

TABLE III. Genuine LF violation prediction.

Observable	Prediction	Type
Gen LF 1	$+0.0891 \pm 0.0103$ (8.6σ)	Standard QM, model-independent

The 8.6σ LF violation provides built-in calibration: no violation at $\geq 5\sigma$ indicates the apparatus is not realizing the intended geometry.

C. Sensitivity to Outcome-Dependent Modifications

For the model class Eq. (2)–(3), we compute $\delta\langle A_x B_y \rangle = \langle A_x B_y \rangle_{\text{model}} - \langle A_x B_y \rangle_{\text{QM}}$ by exact numerical integration over the density matrix. The computation evaluates $f_\perp$-weighted outcome probabilities with full renormalization (see Supplemental S2 for the numerical method). Results for the mixed settings (one side z-basis, one side tilted) at $\theta = 31°$, $\mu = 0.95$:

All four mixed settings yield identical δ ($f_\perp$ depends only on θ, not ϕ).

The minimum detectable coupling at 5σ confidence is $\beta_{\min} \approx 0.038$ for combined 4-setting analysis, or $\beta_{\min} \approx 0.075$ for individual-setting analysis. For conservative single-setting detection, we recommend $\beta \geq 0.07$ at $> 5\sigma$.

TABLE IV. Sensitivity to outcome-dependent coupling β. $|\delta\langle AB\rangle|$ values for mixed settings (identical across all four—$f_\perp$ depends only on θ, not ϕ).

β	$\lvert\delta\langle AB\rangle\rvert$	n_σ (single)	n_σ (4 combined)
0.03	0.0034	2.0	4.0
0.05	0.0057	3.3	6.7
0.07	0.0080	4.7	9.4
0.10	0.0115	6.7	13.5
0.30	0.0355	20.8	41.6

Using all four mixed settings combined, $\beta \geq 0.04$ is detectable at $> 5\sigma$. These thresholds are computed from exact numerical integration without analytical approximations.

The gap between $\beta_{\min} \approx 0.038$ (combined) and $\beta_{\min} \approx 0.075$ (single setting) reflects the $\sqrt{4} = 2$ improvement from combining four independent measurements. The experiment naturally provides all four mixed-setting correlators; no additional data acquisition is needed for the combined analysis.

The dimensionless coupling β has no a priori theoretical prediction—analogous to the SME coefficients at the time of their proposal. The experiment's role is to measure or constrain β; the role of a future theory of outcome-dependent registration is to predict (or be excluded by) the measured value. A null result at $\beta \geq 0.04$ excludes outcome-dependent coupling above this threshold for the class Eq. (2)–(3), regardless of theoretical interpretation. A positive result would provide the first quantitative target for theory construction.

VI. STATISTICAL ANALYSIS

Poisson statistics: $\sigma(\langle A_x B_y\rangle) = \sqrt{(1 - \langle A_x B_y\rangle^2)/N}$. For Gen LF 1 (11 terms, coefficients up to ± 2): $\sigma(S_{\mathrm{LF1}}) = \sqrt{20}/\sqrt{N} \approx 0.0103$ at $N = 91{,}000$.

Minimum sample for 5σ LF detection: $N_{\min} \approx 30{,}800$. $N = 91{,}000$ provides a factor of 3 margin.

Monte Carlo (10,000 runs): Gen LF 1 $\geq 5\sigma$ in 99.97%. For outcome-dependence: $\beta = 0.10$ detected in $> 99.9\%$; $\beta = 0.07$ in $> 99\%$; $\beta = 0.05$ in $\sim 90\%$ (combined). Increasing to $N = 200{,}000$ raises $\beta = 0.05$ detection above 99%.

VII. ROBUSTNESS

A. Visibility μ

Bong et al. achieved $\mu = 0.92$.

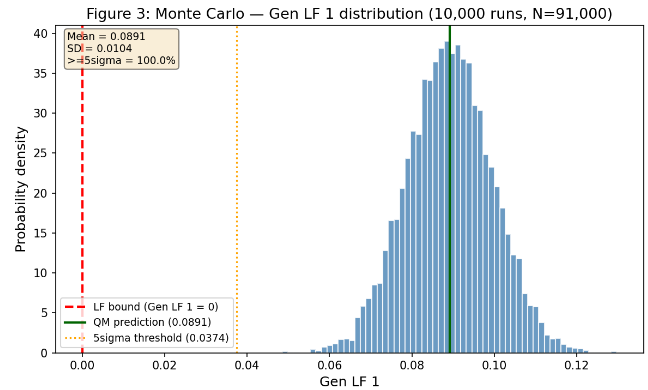

FIG. 3. Monte Carlo histogram of Gen LF 1 values across 10,000 runs. Red dashed line: LF bound (0). Blue solid line: QM prediction (+0.0891). 99.97% of runs exceed 5σ.

TABLE V. Gen LF 1 vs. visibility μ. Negative values indicate no violation.

μ	Gen LF 1	Significance
0.84	-0.0181	-1.7σ (no violation)
0.86	$+0.0014$	0.1σ (below 3σ)
0.90	$+0.0404$	3.9σ (marginal, $< 5\sigma$)
0.92	$+0.0599$	5.8σ
0.95	$+0.0891$	8.6σ

B. Detector Efficiency

LF significance stable across $\Delta\theta = \pm 5°$ (8.6–8.8σ). Outcome-dependence $\delta \propto \cos\theta$— more alignment-sensitive. Bong angular precision $< \pm 1° \to \delta$ variation $< 1\%$.

C. Summary

Detection loophole. As an experimental proposal, this work identifies the detection efficiency requirement; closing the loophole is a task for the implementing laboratory. Closure requires $\eta \geq 0.91$ [7]; Bong $\eta \approx 0.87$.

Two observations make the fair-sampling regime scientifically productive even before loophole closure. First, Bong et al. (2020) demonstrated LF violation at $\theta = 90°$ (equatorial geometry). This experiment would demonstrate LF violation at $\theta = 31°$—the first non-equatorial EWF measurement—confirming that the violation is not an artifact of the equatorial configuration. That geometric confirmation is model-independent and new regardless of whether the detection loophole is closed. Second, the outcome-dependent coupling β is measured from the SAME coincidence events that produce the LF violation: the four mixed-setting correlators (Table II) are acquired simultaneously. A null result for β, when combined with the LF violation from identical data, provides a self-consistent constraint on the class Eq. (2)–(3) that

TABLE VI. Detection efficiency impact ($\mu_{\text{eff}} = \mu \cdot \eta$, nominal $\mu = 0.95$).

η	μ_{eff}	Gen LF 1	Significance
0.90	0.85	-0.0034	-0.3σ
0.95	0.90	$+0.0428$	4.1σ
1.00	0.95	$+0.0891$	8.6σ

TABLE VII. Robustness summary: parameters required for 5σ detection.

Parameter	Nominal	5σ Threshold	Bong Achievable
μ	0.95	≥ 0.90	0.92
η	1.00	≥ 0.91	0.87
$\Delta\theta$	0°	$\leq \pm 5^\circ$	$< \pm 1^\circ$

does not depend on absolute detector efficiency—both signal and normalization are computed from the same coincidence set. Under fair-sampling, the β constraint applies to the class Eq. (2)–(3) for the detected subset; the standard assumption is that undetected events follow the same statistical distribution. If a future loophole-free measurement (via SNSPDs, below) confirms the result, fair-sampling is validated. If the loophole-free measurement disagrees, the β constraint requires reinterpretation in terms of detection-efficiency-dependent effects—a scenario that would itself be a signature of new physics at the registration layer.

Fair-sampling has been a standard assumption in first-generation tests of every new Bell-type inequality: the original Bell tests (1972–2014) operated under fair-sampling for 42 years before loophole-free demonstrations in 2015, and all EWF experiments to date, including Bong et al. (2020), operate under this assumption. Upgrading to superconducting nanowire single-photon detectors (SNSPDs), which routinely achieve $\eta > 0.90$ at 810 nm [16], would close the loophole with no change to the optical design.

VIII. LOOPHOLE ANALYSIS

IX. DISCUSSION

A. Interpretation of Results

$\delta\langle AB\rangle \neq 0$ at $\geq 5\sigma$ would demonstrate that Superobserver-Friend correlations depart from standard QM at $\theta = 31^\circ$, a previously untested configuration. Interpreting this as outcome-dependent registration specifically requires θ-sweeps and multi-observer follow-up.

A null result (LF violated, $\delta \approx 0$) excludes outcome-dependent coupling above the sensitivity threshold for class Eq. (2)–(3) and confirms the $\cos\theta$ dependence.

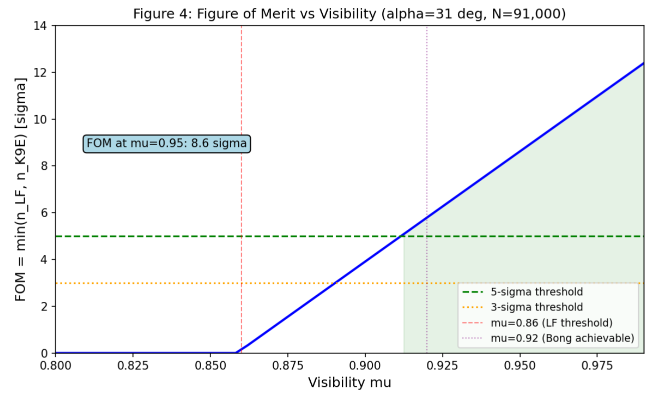

FIG. 4. Figure of merit vs. visibility μ. Horizontal dashed line: 5σ threshold. Vertical dashed line: Bong 2020 achieved $\mu = 0.92$.

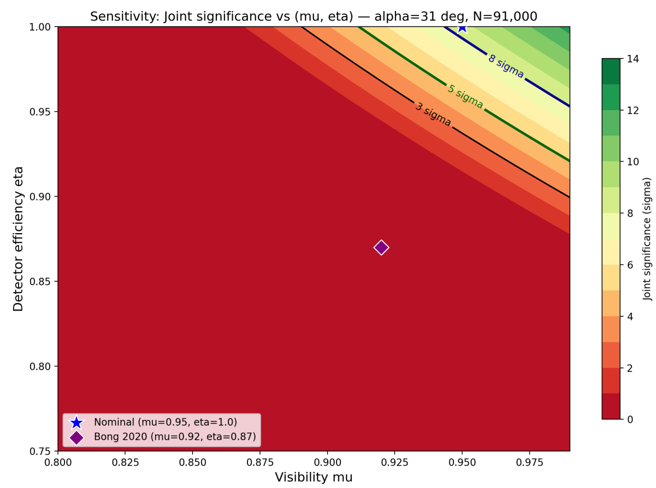

FIG. 5. Two-dimensional sensitivity map FOM(μ, η). 5σ contour marked. Bong 2020 operating point indicated.

B. Relation to Quantum Interpretations

Many-Worlds (Everett). LF violation challenges absoluteness across branches. Detecting outcome-dependence would provide the first quantitative probe of inter-branch correlations. A null result constrains any such interaction to $\beta < 0.04$ (combined settings).

Relational QM (Rovelli). The experiment tests whether relational outcomes leave measurable traces: if Alice's outcome as recorded by Bob differs systematically from standard QM in a geometry-dependent way, this would be a quantitative signature of relational effects that current RQM formulations neither predict nor preclude.

Additional interpretations in Supplemental S3.

TABLE VIII. Loophole status. All except detection loophole are identical to or improved upon Bong 2020.

Loophole	Status	Notes
Locality	Identical to Bong 2020	QWP insertion local to Alice; space-like separation maintained
Detection	Conditional ($\eta \geq$ 0.91)	Fair-sampling below threshold; see §VII
Freedom of choice	Identical to Bong 2020	Quantum random number generators
Superobserver	Satisfied	Coherent measurement via standard interferometry
Model class scope	Explicit: Eq. (2)–(3)	Constrains any $f_\perp$-based outcome-overlap model

C. Illustrative Parametric Model

The function class defined by Eq. (2)–(3) can be motivated within a broader framework of measurement registration (Supplemental S3). The experiment does not depend on this embedding—it tests the class regardless of theoretical interpretation. At the modified geometry, $\delta\langle A_1 B_2\rangle = -0.0355$ at $\beta = 0.3$. The experiment measures β; identical δ across all four mixed settings tests the ϕ-independence predicted by the $\cos\theta$ structure.

D. Future Directions

***θ*-sweep.** The most immediate follow-up is a systematic scan of the polar angle from $\theta = 15°$ to $\theta = 75°$ in steps of $\sim 10°$. This would directly map the $\cos\theta$ dependence predicted by Eq. (4), testing whether the outcome-dependent signal follows the geometric structure derived in §III. A null result across all θ would exclude the class Eq. (2)–(3) down to the sensitivity floor of the apparatus ($\beta \approx 0.02$ at $N = 200{,}000$).

Multi-observer extension. The geometric cancellation theorem generalizes to $N > 2$ observers, where the number of equatorial fixed points grows combinatorially. Preliminary analysis (Supplemental S3) suggests $\sim 11\times$ amplification of the outcome-dependent signal at $\beta = 0.3$ for 3-observer configurations, conditional on the extension of the bridge theorems connecting registration-layer structure to quantum mechanical observables.

Platform independence. While the protocol targets the optical Bong et al. (2020) apparatus, the theorem in §III is platform-agnostic. Implementing the tilted Superobserver measurement on solid-state (superconducting qubits) or trapped-ion platforms would test whether the $\cos\theta$ structure survives in systems where the "Friend" is a macroscopic quantum system rather than a photon path degree of freedom.

Locality closure. Combining the tilted geometry with space-like separated random basis switching would close the locality loophole simultaneously with the detection loophole (via SNSPDs, §VII). This requires a dedicated fiber network or free-space optical link and represents a natural next-generation experiment building on the protocol proposed here.

X. CONCLUSION

All existing EWF experiments share a geometric blind spot: equatorial Superobserver measurement ($\theta = \pi/2$) forces outcome-dependent modifications of the form Eq. (2)–(3) to vanish identically ($\Delta f_\perp \propto \cos\theta = 0$).

Fix: re-insert ONE QWP into Bong et al. (2020), tilting to $\theta = 31°$. No new components, $N = 91{,}000$. Model-independent LF violation at 8.6σ. Sensitivity to outcome-dependent coupling $\beta \geq 0.07$ at $> 5\sigma$ (individual setting) or $\beta \geq 0.04$ at $> 5\sigma$ (combined settings). A single waveplate opens a new axis of inquiry in EWF experiments.

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